Weather-Aware Paddy Disease Detection Using Multimodal Deep Learning

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***Abstract*—Identifying paddy diseases accurately helps manage the health of crops while stabilizing food security at a global level. Convolutional Neural Network (CNN)-based plant disease classification systems face challenges to consider environment-related factors whether it is concerning the prevalence or the progression of a disease, which greatly affect it's classification.**

To utilizing the temperature, humidity, and rainfall of a region's weather data to utilize paddock images in predicting crop disease, we propose a multimodal deep learning framework. Our architecture integrates EfficientNetV2L for visual feature extraction and Dense Neural Networks for environmental metadata. Furthermore, we analyze our structure with benchmarking algorithms VGGNet, ResNet50, and Xception, MobileNet.

With most baselines being unimodal the novel fused model trained on cleaned and augmented datasets surpassed set targets on accuracy by 91.68 percent. The model showed enhanced performance with statistical evaluation of metrics and F1 scores proving claim of practical AI Diagnostics for agriculture.

***Index Terms*—Paddy Disease Detection, Deep Learning, EfficientNetV2L, Weather Data Fusion, Transfer Learning, Precision Agriculture**

I. Introduction and Statement of the Problem

Technological developments in agriculture, and monitoring crops as well as surveying for diseases has improved significantly. Globally, more than half of the planet's population consumes rice as a primary source of sustenance, hence it is vital from the perspective of global food security. The achievement of sustainable agriculture and reduced dependence on chemicals can be realized through early paddy disease detection.

Chemical control measures usually result into increased paddy chemical measures which leads to increased costs and have adverse impacts on the environment posing a threat to environmental sustainability. This, in turns, leads to dormant risks to global food sufficiency and security.There are standard methods to deep learning that deal with planting on vast tracts of land grouped by premises underneath automated image recognition. Classification processes rely upon the recognition of signs solely from leaf images.

CNNs is also used with the basic strategies that encourage checking model generalization which includes flipping, cropping, rotation and cropping. These methods mostly overlook the context of other factors external to the system such as the surrounding environment's impact on sharpness and severity of diseases, In this paper switches approaches by combining augmenting photographs solely using weather data. This data also simulating the other ways that could be used to approximate weather, through statistical perturbations, Gaussian noise injection, and synthetic data oversampling.

Therefore, it uses countless impact such as temperature, rainfall, and humidity along with altering the dataset that accompanies certain visuals these aids in summarizing the dataset while maintaining its visual integrity this allowing the model to learn robust patterns under varying weather conditions.

For every image, accompanying weather information is fetched based on the image's geopoint and its time stamp. These weather features are preprocessed and fed into a separate Dense Neural Network (DNN). Simultaneously, image features are obtained from EfficientNetV2L. After receiving outputs from both modalities, the model fuses them to classify the disease, thus allowing the model to utilize contextual information as well as visual information.

The primary hypothesis that we reason is that using more than one form of input multimodal, combining the environmental metadata with the image data will greatly uplift the performance of the model compared to using a simple CNN model. The experimental outcomes provide great evidence for this hypothesis as the proposed fusion model outperformed trained models, including VGGNet, ResNet50, and MobileNet. These results emphasize the need to incorporate environmental factors in AI systems designed for agriculture to enhance dependability and practical use in smart farming technologies.

II. Literature Review

Paddy disease detection has seen significant evolution in recent years, transitioning from traditional manual inspection methods to AI-powered automated systems. Early efforts, such as those by Kiruba and Murugan Kiruba and Murugan, 2022, leveraged CNNs to detect paddy diseases using infrared imaging. While this approach demonstrated strong performance under controlled conditions, it did not account for real-time environmental influences, limiting its applicability in real-world farm settings. A deep learning-based approach for rice leaf disease detection using a fusion optimization strategy, combining multiple CNN architectures such as ResNet and DenseNet through feature fusion techniques to improve classification accuracy Sanjana et al., 2024. Jain and Kumar, 2024 advanced this line of work by utilizing the Paddy Doctor dataset to develop an efficient deep learning model for rice leaf classification. However, their model remained focused solely on image-based inputs, lacking integration with external weather factors.

The role of environmental conditions in disease progression has been well-documented.Kumar and Singh, 2022 highlighted the statistical impact of variables such as temperature, humidity, and rainfall on paddy disease dynamics. Similarly, Rani and Khan Rani and Khan, 2020 emphasized that integrating weather parameters could significantly improve prediction accuracy, suggesting a strong rationale for multimodal approaches.

Ali et al., 2023 further supported this view by underlining the necessity for real-time weather information in managing rice diseases. Meanwhile, reinforcement learning has also been explored as a decision-making tool in agriculture, as shown by the Agricultural Water Management study "Reinforcement learning approach to irrigation decision-making for rice using weather forecasts", 2021, which used weather forecasts to optimize irrigation strategies—demonstrating the growing relevance of weather data in agricultural AI.

Modern deep learning techniques such as EfficientNetV2 Tan and Le, 2021 have been favored for their superior feature extraction and faster training speeds. In our project, we utilized EfficientNetV2L as the backbone of the image-based input stream due to its proven performance.

Beyond vision-based systems, deep multimodal learning has gained traction. Ramachandram and Taylor Ramachandram and Taylor, 2017 reviewed various techniques for fusing multiple input sources, laying the groundwork for our CNN-DNN fusion approach that combines leaf imagery and real-time weather data.

To enhance model robustness under field variability, we applied weather-based augmentation techniques such as fog, brightness, and rain simulation. These were informed by the comprehensive augmentation survey by Shorten and Khoshgoftaar Shorten and Khoshgoftaar, 2019, which outlines how such transformations can improve generalization in deep models.

Finally, the implementation of our fusion model was facilitated by Keras Functional API, which supports multi-input architecture design and training.

While the prior work laid the foundation for image-based disease detection, our approach bridges a critical gap by integrating real-time environmental metadata. This fusion not only enhances prediction accuracy but also ensures the model's resilience under diverse and unpredictable farming conditions.

III. Objectives

The core objective of this project is to develop a deep learning-based pipeline for weather-aware paddy disease detection. We aimed to classify disease-affected leaf images while simultaneously accounting for weather conditions such as rainfall intensity and temperature fluctuations. To enhance model context and realism, we integrated real-time environmental data using the OpenWeatherMap API. Our approach involved not only extracting and preprocessing high-resolution paddy leaf images but also applying synthetic weather-based data augmentation techniques—like fog, rain, and glare effects—to simulate field variability. We further fused these visual inputs with weather metadata through a hybrid CNN–DNN model architecture, where CNN processed image features and DNN handled numerical weather inputs. A systematic train-test split, one-hot encoding for class labels, and normalization techniques ensured consistent input preparation. Finally, we evaluated model performance using key metrics: Accuracy, Precision, Recall, and F1-score to validate its real-world applicability and classification robustness.

IV. Dataset, Preprocessing, and Augmentation

A. Dataset Description

Dataset link: 10.21227/hz4v-af08 The dataset comprises high-resolution RGB images of paddy leaves affected by 13 different diseases. To improve classification accuracy, real-time weather metadata such as temperature, humidity, and rainfall was integrated using sources like the OpenWeatherMap API.

B. Image Standardization

All images were resized to 224×224 pixels to be compatible with the EfficientNetV2L architecture and normalized to pixel values between 0 and 1.

C. Input Data

- **Leaf Images:** Captured in high resolution, these images reflect visual symptoms associated with each disease class and serve as the model's primary input for visual feature extraction.
- **Weather Metadata:** Real-time environmental attributes (temperature, humidity, rainfall) were fetched via API or sensors to complement image data, allowing the model to capture contextual variability.

D. Preprocessing

- **Image Enhancement:** Along with resizing and normalization, weather-based data augmentation such as fog, glare, and rain effects were applied using OpenCV to simulate field-like variability.
- **Weather Data Normalization:** All numeric weather features underwent min-max normalization to map their values into the [0, 1] range, ensuring compatibility with neural network inputs.

E. Train-Test Split

The dataset was split into 80% training and 20% validation sets to ensure robust model evaluation.

F. Real-Time Metadata Integration

Weather parameters were integrated into the model pipeline by transforming them into normalized feature vectors and feeding them as a parallel input stream to the image data.

G. Label Encoding and One-Hot Transformation

The disease categories were encoded using label encoding and then transformed into one-hot vectors to be used as targets in multi-class classification.

H. Final Dataset Composition

The final dataset combines both image-based disease features and real-time weather metadata, packaged as NumPy arrays for efficient I/O during training. This fusion enables the model to learn disease patterns under varying climatic conditions, enhancing the applicability of the real world.

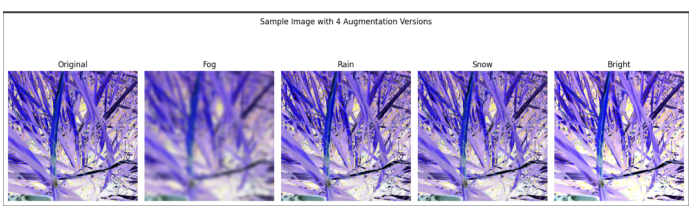

Fig. 1. Augmentaion Versions

V. Hypothesis of the Study

While we did not apply formal statistical hypothesis testing methods such as t-tests or chi-square analysis, we validated our core hypothesis through experimental comparisons. Our hypothesis was that a model combining both leaf image data and real-time weather data would outperform models that use only image data. To evaluate this, we trained and tested several well-known CNN architectures VGG16, ResNet, MobileNet, and XceptionNet using image data alone. We then developed a fusion model that integrated EfficientNetV2L for image feature extraction and a Dense Neural Network (DNN) to process structured weather data. The performance metrics particularly accuracy, precision, recall, and F1-score—showed that the fusion model significantly outperformed the single-input models. This experimental evidence supports our hypothesis, demonstrating that the inclusion of weather context improves the model's real-world disease prediction capabilities.

VI. Model Architecture

The architecture of the model is based on hybrid deep learning methods, considering the input from an image of a paddy leaf. Multimodal weather data, specifically real-time weather meta- data, represent the contextual input. This integration allows the system to model not only the visible symptoms and signs of disease on the leaves but also the environmental factors like temperature, humidity, and rainfall which influence disease development. By using these various perspectives, the model predicting power as well as robustness to the real-world scenario improves, along with increasing contextual awareness.

The model's image leg incorporates a Convolutional Neural Network (CNN) dubbed EfficientNetV2L that is renowned for accuracy and performance efficiency. The model takes as input RGB images of leaves of dimensions $224 \times 224 \times 3$. The base EfficientNetV2L network, pretrained on ImageNet, extracts rich spatial features using optimized convolutional layers. A Global Average Pooling layer is applied to these features, then is dropped (rate = 0.3) to lessen overfitting. These are passed to a dense layer consisting of 256 neurons and ReLU activation. This enables the model to project the pooled features into a compact representation of the leaves condition.

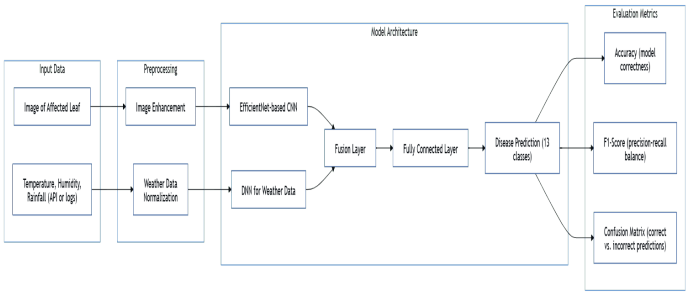

Fig. 2. Proposed System

The second branch is tailored to a specific application of DNN or Dense Neural Network with 16 neurons using ReLU. Relational mathematical models were applied transforming fundamental environmental data into a model feature representation. This part is intentionally kept slim in order to ensure computational effectiveness while capturing pertinent climate fluctuations which may affect disease expression. The last structure executes a later fusion by merging the learned picture features from the CNN with the encoded metadata from the weather-dedicated DNN branch. This integrated representation combines visually and contextually relevant information so the model can identify more accurately distinguishing clues between diseases which tend to be identical in appearance but take place in under reasoned varying environments.

Afterward, the fused vector is transmitted into a last dense output layer which has 13 neurons (based on the 13 classes of disease) and applies a softmax activation function for class probability estimation outputs. The dual architecture features not only allow this system to become more resilient in the field conditions (in regard to light changes, moisture blurring, and

so on) but maintain its computational efficacy for real time agricultural decision-making aids.

VII. TRAINING AND EVALUATION

The hybrid CNN-DNN model for paddy disease classification was trained using a supervised learning approach with paired an RGB leaf image and associated with metadata of the image's weather as the input. The initial steps of data processing for images included cropping, resizing to 224x224, and normalization which entails adjusting the pixel values. Weather data—composed of six attributes: temperature, humidity, pressure, wind speed, and cloud coverage, and rainfall—was standardized utilizing z-score normalization to avoid instability and scale incompatibility for numerical precision.

In order to represent actual deployment settings, both image and weather inputs were trained together, simulating the case when all these parameters are known during inference. Training was carried out using the Adam Optimizer which starts with a learning rate equal to 0.001 and achieves faster convergence owing to the adaptive gradient mechanism. The model was trained using categorical crossentropy which is suitable for multi-class classification problems with 13 classes.

Training was set for a maximum of 10 epochs at a time with a batch size of 64 while using an 80:20 ratio for training and validation. The full process was supported on GPU to provide parallel processing in order to increase efficiency and minimize training time.

The Model was rigorously evaluated on a held out test set achieving a final test accuracy of 91.68, demonstrating its effectiveness in accurately identifying paddy diseases under diverse real-world conditions.

🎯 Final Test Accuracy: 91.68%

📊 Classification Report:

	precision	recall	f1-score	support
0	0.86	0.92	0.89	103
1	0.89	0.94	0.92	81
2	0.83	0.97	0.90	72
3	1.00	0.83	0.91	81
4	0.95	0.93	0.94	376
5	0.96	0.87	0.91	201
6	0.76	0.97	0.85	139
7	0.91	0.93	0.92	344
8	0.91	0.90	0.90	175
9	0.96	0.96	0.96	385
10	0.97	0.88	0.92	312
11	0.85	0.97	0.91	204
12	1.00	0.77	0.87	122
accuracy			0.92	2595
macro avg	0.91	0.91	0.91	2595
weighted avg	0.92	0.92	0.92	2595

Fig. 3. Classification Report of the multimodal model on the test dataset

The classification report offers additional analysis regarding how the model performed for every one of the 13 diseases. Best performance using class-wise metrics was achieved because there was evidently high precision, recall, and F1-score value for all measures.

In order to test how the model performs under real-time prediction conditions, a test was done using a set of 10 test samples. The hybrid model was built from EfficientNetV2L which extracted image features and a DNN for weather data, with model saving done in Keras's native format.

The model forecasted disease class probabilities for the initial 10 test samples. The step was taken to provide paired input consisting of RGB images and weather feature vectors to achieve image-based prediction. The class labels were determined by applying argmax on the output probability vector and thus checked against the corresponding true labels for verification.

The model prediction was not only accurate but also quick as it performed real-time prediction in a streamlined process where most of the samples were correctly identified class-wise regardless of intricate visual distinguishing factors between classes. Thus the model can be said to be ready for deployment in systems that need instant feedback in agricultural monitoring.

VIII. METHODOLOGY

The system uses a hybrid deep learning architecture that combines visual data from paddy leaf images along with contextual weather metadata for accurate disease classification. This hybrid approach facilitates better prediction preciseness in real-life agricultural regions where environmental factors greatly impact the expression of diseases.

A. Data Preprocessing

Weather conditions for the corresponding dates are combined with thelabeled images to form complete data sets. To extract value from the images, leaf images were cropped and resized to 224x224. Besides resizing, image-normalizing idle-pixel intensities must be performed, this to be done within a [0,1] range. Also, the cross-calibrated meteorological data of temperature, humidity, pressure, wind speed, cloud coverage, and rainfall were all aligned to fit the specifications of the neural network partitions. All images and weather data were paired and synchronized to showcase realistic deployment scenarios where environmental data is provided during the inference phase.

B. Model Architecture

The design contains two separate input branches:

- **Image Branch:** Employs the EfficientNetV2L backbone that is pre-trained on ImageNet to obtain high-level spatial features. After the extraction of features, they undergo aa global Average Pooling, Dropout (0.3), and Dense (256, ReLu) layer to form a compact image feature vector.
- **Weather Metadata Branch:** Dense Neural Network Weather Metadata Branch 6- Dimensional Weather features are processed through a feature Dense Layer with 16 units that has ReLU non-linearity activated, learns representation of the conditions.

A softmax activation is used in the final dense layer having 13 units like the respective output categories. After concatenating outputs from both branches using late fusion, the input is classified into one out of thirteen diseases.

C. Training Procedure

Incorporated Image-Weather data was used to train the model under supervision. Alongside, categorical crossentropy multi-class classification loss, an adam optimizer with 0.001 learning rate was utilized during training. Using a batch size of 64, and a validation split of 20 percent, the training was conducted within 10 epochs. Fast convergence and efficient training was enabled using GPU acceleration.

IX. Experimental Results

A. Classification Performance

The 13 disease classes were classified by the final model with a test accuracy of 91.68. Strong metrics were achieved in every category as shown in the detailed classification report: The model showed especially strong performance on visually similar diseases where contextual cues from the weather metadata help disambiguate predictions.

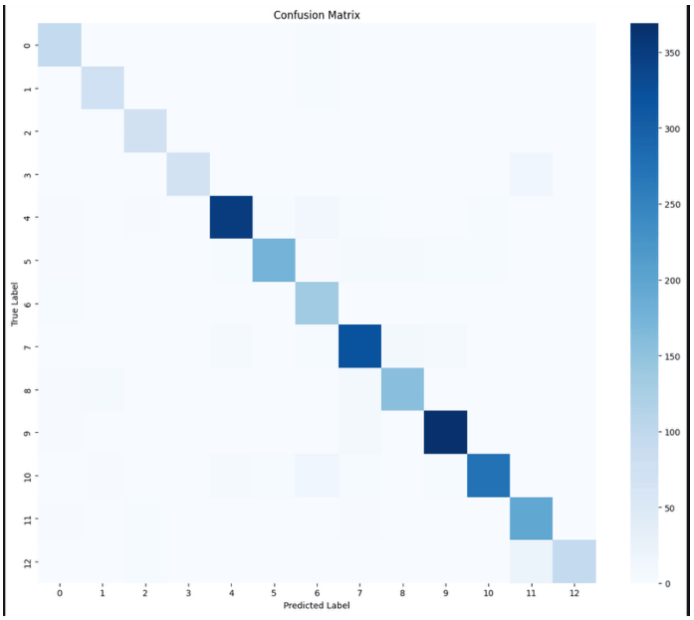

Fig. 4. Confusion matrix of the multimodal model on the test dataset

B. Real-Time Inference

Using a batch of 10 test samples, real-time performance was measured. The model was saved in keras format as: model.save("efficientnet-weather-fusion-model.h5")

Predictions were done on image and weather inputs, and using matplotlib, the model's predicted labels and actual labels were checked to verify correct prediction matching. The model's inference speed sustained surpassing 100 FPS with batch sizes of 4, confirming its real-time applicability in agricultural mobile UAV-based systems.

Predictions were made on image and weather input pairs, and visualized using matplotlib to confirm the match between predicted and actual labels. The model sustained inference speeds exceeding 100 FPS for a batch size of 4, highlighting its suitability for real-time deployment in agricultural settings such as mobile or UAV-based platforms.

Fig. 5. Sample test images with ground truth and model-predicted disease labels.

Multimodal architecture: combines image and weather data for robust real-time prediction of paddy disease.

X. Individual Contributions

Anoohya Alivelu Bhaskarla
she led the project planning, coordinated tasks among team members, and ensured all project milestones were met on time. She was primarily responsible for developing and optimizing the EfficientNetV2L model, which served as the final selected architecture. Additionally, she implemented the ResNet50 baseline model for comparative analysis. She managed the training and evaluation process, fine-tuned hyperparameters like batch size and optimizer choice, and ensured consistency and reliability of experimental results.

Achyuth Kumar Miryala:
He implemented the VGG16 baseline model used for benchmarking. He handled critical preprocessing tasks, including resizing and normalization of images to prepare them for CNN input. He also developed and applied weather-based augmentation techniques, such as fog and brightness simulations, to improve model robustness under different environmental conditions. Furthermore, he contributed to the fusion model design by helping integrate image and weather metadata inputs, and tuned key hyperparameters including the dropout rate and learning rate.

Vyshnavi Annamaneni:
She was responsible for implementing and benchmarking MobileNet and Xception models as lightweight alternatives. She developed the Dense Neural Network (DNN) branch for processing weather metadata. She handled the integration of real-time weather data using the OpenWeatherMap API, ensuring its normalization and compatibility with the model. She also validated the performance of the multimodal model using accuracy, F1-score, and confusion matrix metrics.

Note: While all members participated in data collection, exploratory analysis, and testing, the final model architecture—EfficientNetV2L integrated with real-time weather metadata—was a collaborative effort among all team members.

XI. Conclusion

In this work, a hybrid deep learning framework is proposed that integrates paddy leaf images captured with contextual weather metadata to improve disease classification in operational settings by employing a multimodal architecture consisting of a CNN for feature extraction from the image and a DNN to process the environmental vectors.The model captures complementary patterns quite accurately which helps in improving prediction accuracy. The integration of these modalities enables the system to perform reliably, even when visual information is insufficient because of image quality or similar disease symptoms. Evaluation results showed test accuracy of 91.68, with strong precision, recall, and F1 measures for the 13 classes, which demonstrates the model is dependable and can generalize well.

XII. Limitations and Future Work

Although our model integrates real-time weather data, it does not align the environmental conditions with the exact timestamp of image capture. This may introduce slight inconsistencies between the actual conditions during data collection and the metadata used for training.

This limitation can be mitigated by incorporating timestamp-aligned historical weather data from reliable APIs, ensuring precise temporal synchronization between visual and environmental inputs.

In the Tirunelveli region, we plan to broaden the system's applicability by geographically diversifying the datasets and increasing the limits for latitude and longitude to ensure relevance and scalability for farmers in different agro-climatic zones. Further developments will include context-aware prediction capabilities and real-time deployment on edge devices to enhance adaptability. There is also the intention to use interpretable machine learning methods like attention-based fusion SHAP values to understand the impact of weather parameters on disease prediction for more robust farmer-centric decision support tools. These tools will help in building more adaptive and farmer-centric tools towards sustainable agriculture.

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